

# Procedure

## 1. Identify Threats:

- a. Begin by compiling a comprehensive and detailed list of all potential threats that could potentially impact the organization's systems and data. This includes both internal and external threats such as cyber attacks, data breaches, hardware failures, natural disasters, and human errors. Consider consulting with various departments to ensure no threats are overlooked, and periodically update this list as new threats emerge or the organization's infrastructure evolves.

## 2. Assign DREAD Scores:

- a. For each identified threat, conduct a thorough evaluation and assign a score ranging from 0 to 10 in each of the five DREAD categories: Damage Potential, Reproducibility, Exploitability, Affected Users, and Discoverability. This process should involve leveraging expert judgment, historical data, and any available threat intelligence to inform the scores accurately. The aim is to produce an objective assessment of each threat's potential impact on the organization.

## 3. Calculate Total DREAD Score:

- a. Once scores have been assigned in each category, calculate the total DREAD score for every threat by summing the individual scores from the five categories. This total score will have a potential maximum value of 50. This step provides a quantifiable measure of the overall severity and urgency of each threat, allowing for straightforward comparisons between them.

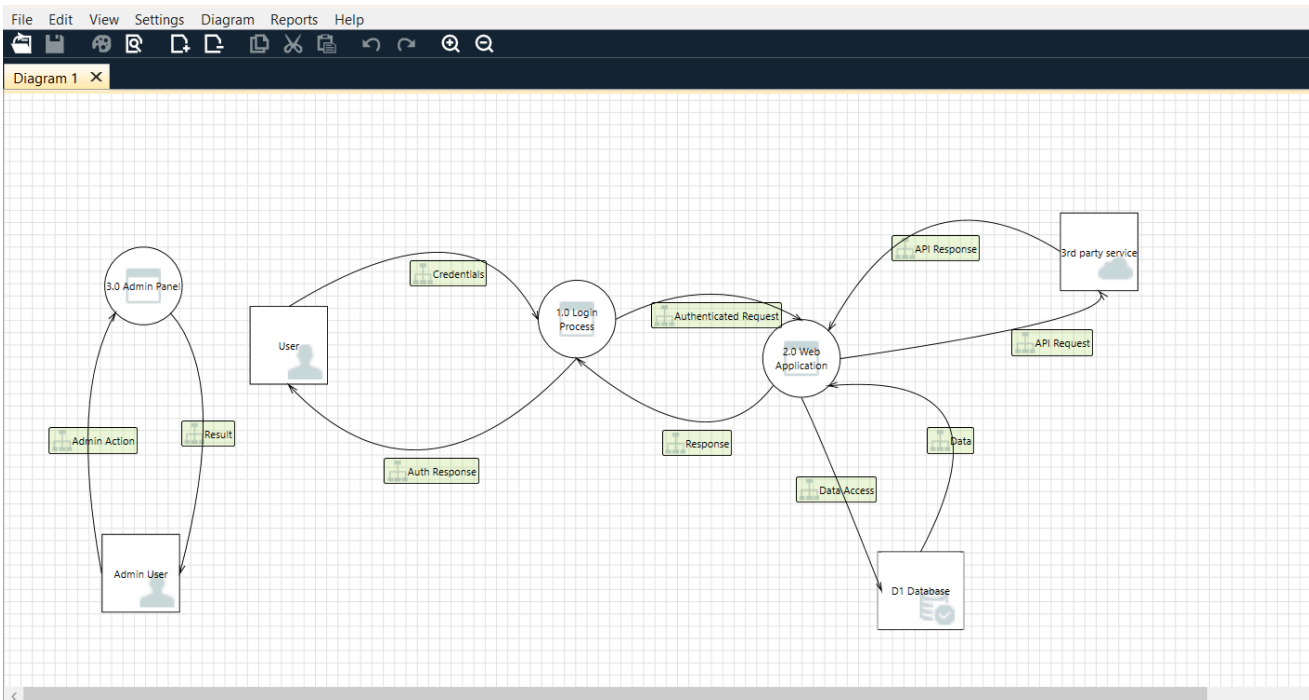
## 4. Prioritize Threats:

- a. With the total DREAD scores calculated, proceed to rank the threats in order of priority. Higher scores correlate with a higher priority, indicating that these threats should be addressed more urgently in mitigation efforts. This prioritization helps allocate resources effectively and ensures that the organization addresses the most critical vulnerabilities first.

## 5. Document Findings:

- a. Create a detailed and organized table to document each threat alongside its corresponding DREAD scores and total score. This table should be comprehensive, easy to read, and regularly updated to reflect any changes in threat assessments or new developments. It serves as a crucial reference for decision-makers and stakeholders, aiding in the strategic planning and implementation of security measures.

# Output



| Threat Description       | Damage Potential | Repro ducibi lity | Exploita bility | Affected Users | Discover ability | Total DREAD Score | Priority |
|--------------------------|------------------|-------------------|-----------------|----------------|------------------|-------------------|----------|
| SQL Injection Attack     | 10               | 9                 | 8               | 9              | 8                | 8.8/10            | Critical |
| Phishing Scam            | 8                | 9                 | 7               | 8              | 9                | 8.2/10            | High     |
| Denial of Service Attack | 8                | 9                 | 8               | 9              | 8                | 8.4/10            | High     |

## Calculations

1. SQL Injection

$$\frac{10 + 9 + 8 + 9 + 8}{5} = \frac{44}{5} = 8.8$$

2. Phishing

$$\frac{8 + 9 + 7 + 8 + 9}{5} = \frac{41}{5} = 8.2$$

3. DoS

$$\frac{8 + 9 + 8 + 9 + 8}{5} = \frac{42}{5} = 8.4$$

## Analysis

- **SQL Injection Attack** scores the highest with a total of 40, indicating it should be a key focus for mitigation.
- **Phishing Scam** has a considerable impact on affected users, suggesting a need for enhanced user education and technical defenses.
- **Denial of Service Attack** also poses significant risks, warranting attention to network security measures.